

Table 1: Experimental dimensions. The suite is synthetic and mechanism-oriented; none of these dimensions should be read as a calibrated city model.

Dimension	Values or construction	Purpose
Network families	Two-link traps, multi-link traps, parallel links, Braess-like networks, synthetic grids, offline heterogeneous grids	Separate hand-designed capacity traps from less favorable sanity checks.
Demand and load	Multiple OD demand levels with maximum-utilization filters at 2 and 5	Distinguish moderate regimes from overload stress tests.
Latency functions	BPR-style latency with fixed α and varied β	Vary convexity of congestion externalities.
Route candidates	Shortest- k , edge-penalty, random-reasonable, and disjointness-preferred route sets	Test whether conclusions depend on one route-generation method.
Recommendation structure	Deterministic, diversified, correlated, partial-compliance, and greedy baselines	Compare shared rankings against diversity and approximate social-cost awareness.

counts, route entropy, route Herfindahl-Hirschman concentration (HHI), route overlap, and PoM relative to the best available diversified baseline.

Evaluation protocol. The reproducibility-critical steps are as follows.

Synthetic sweep protocol

For each random seed, network family, parameter setting, and OD demand, build a directed graph with free-flow times and capacities. Generate one route-candidate set; for each recommendation policy, induce route and edge flows, compute BPR latencies, and record social cost, average latency, maximum utilization, route concentration, route overlap, and warnings. Compare monoculture with the best implemented diversified baseline and export the reported summaries.

Greedy marginal-social-cost baseline

Initialize route flows at zero. Assign each small demand quantum to the candidate route with the lowest incremental social cost under the current edge loads, then update the loads. This is an oracle-style comparison baseline, not a claimed continuous global optimum.

Correlated recommender family

For each OD pair, combine shared and independent perturbation vectors. At $\rho = 0$, rankings are independent across user groups; at $\rho = 1$, all groups receive the same perturbed ranking; intermediate values interpolate between these cases.

Parameter coverage and formula checks. Table 3 records the variables varied or retained for audit. Closed-form monoculture, split, capacity-split, and partial-compliance formulas are checked against the simulator on representative $\beta = 2$, $\beta = 4$, and capacity-stress settings; a small checker grid also confirms beneficial split assignments in the two-link construction.

Table 2: Recommendation policies used in the reported comparisons. The greedy row is a tractable oracle-style benchmark, not a certified optimum.

Policy family	Mechanism	Interpretation
Free-flow monoculture	Send all OD demand to the free-flow-shortest candidate route	Shared deterministic recommendation.
Shared noisy monoculture	Draw one perturbed edge-weight model and send all users to its top route	Correlated ranking with common noise.
Top- k and softmax diversity	Split demand across candidate routes using uniform or cost-weighted shares	Simple recommendation diversity.
Independent multi-model	Draw independent perturbations across user groups	Reduced correlation through model diversity.
Correlated recommender	Interpolate between independent and shared perturbations using ρ	Direct correlation sensitivity.
Partial compliance	A q fraction follows monoculture; the rest uses a diversified fallback	Behavioral uptake sensitivity.
Greedy marginal cost	Sequentially assign flow to the route with lowest current marginal social cost	Approximate social-cost-aware baseline.

Table 3: Recorded parameter coverage. Capacity-trap families deliberately favor the mechanism; the offline heterogeneous grid is not hand-designed as a two-link trap.

Component	Parameters varied or recorded
Capacity-trap networks	Base capacity, shortcut-capacity ratio, free-flow ratio, and number of parallel or candidate links.
Grid networks	Grid size, OD selection, free-flow distribution, capacity distribution, and capacity/free-flow correlation regime.
Latency and demand	BPR α , exponent β , total OD demand, edge utilization, and congested-edge counts.
Route sets	Candidate count, route overlap, distinct-route count, and shortest-to-longest candidate ratio.
Policies	Policy class and parameters, compliance q , correlation ρ , perturbation scale, and fallback policy.

To avoid letting extreme overload dominate the narrative, we report both raw stress-test summaries and filtered moderate-regime summaries. The two main filters bound maximum edge utilization by 2 and by 5. We also inspect harmless regimes where PoM is near 1, and route-set sensitivity across shortest- k , edge-penalty, random-reasonable, and disjointness-preferred route-generation methods.

6 Results

Capacity-trap mechanism. Figure 1 illustrates the basic two-link mechanism. Deterministic free-flow monoculture concentrates all flow on the shortcut. Split policies pay a small free-flow penalty on the slower relief link but reduce the convex congestion term on the